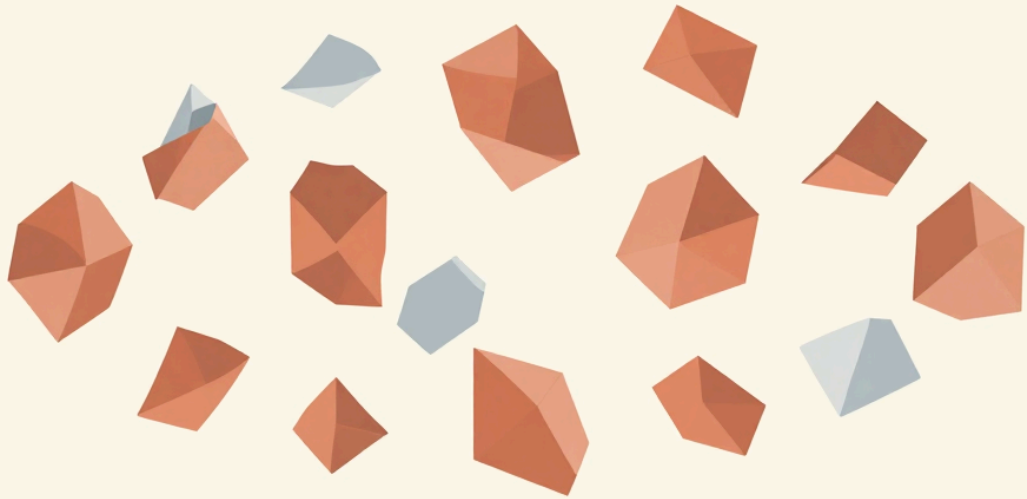




Attention is All You Need

PAPER EXPLAINED

Attention Is All You Need: No Recurrence Required

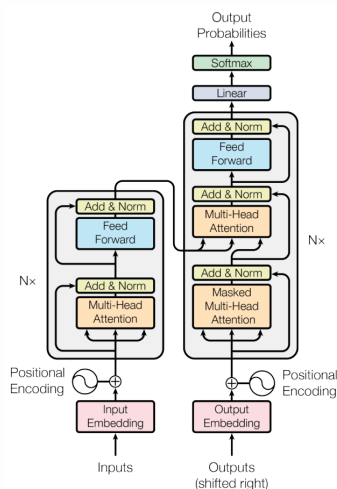
Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit et al.

<https://arxiv.org/abs/1706.03762>

Modeling Dependencies Without Recurrence

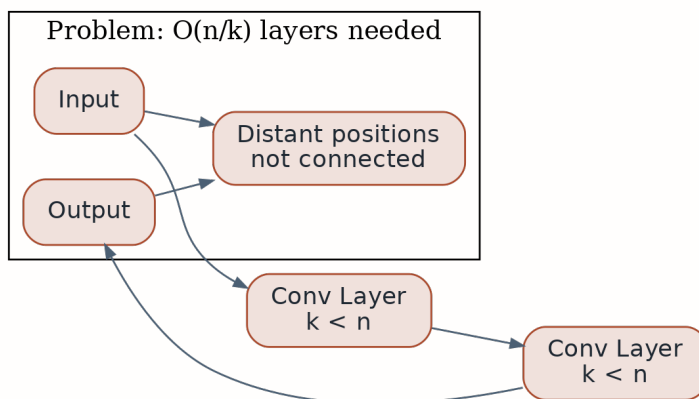
- Attention models dependencies regardless of distance in input or output sequences.
- The Transformer relies entirely on attention, eschewing recurrence to draw global dependencies.
- Future plans include extending the model to images, audio, and video modalities.



The Transformer - model architecture.

Sequential Bottlenecks in Previous Models

- Recurrent models preclude parallelization within training examples.
- Previous convolutional models make it difficult to learn dependencies between distant positions.
- A single convolutional layer does not connect all pairs of input and output positions.

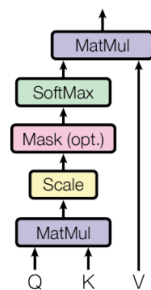


Convolutional models require $O(n/k)$ layers to connect distant positions, creating sequential bottlenecks.

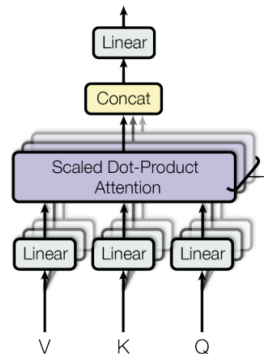
Pure Self-Attention Architecture Design

- The Transformer uses stacked self-attention and point-wise fully connected layers for both encoder and decoder.
- It employs $h=8$ parallel attention heads, each with dimension $dk=dv=64$, to attend to different subspaces simultaneously.
- Self-attention connects all positions with constant sequential operations, unlike recurrent layers requiring $O(n)$ operations.

Scaled Dot-Product Attention



Multi-Head Attention



(left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

State-of-the-Art Translation and Parsing Results

- The big Transformer model achieves a state-of-the-art BLEU score of 28.4 on WMT 2014 English-to-German translation.
- On WMT 2014 English-to-French, the model reaches a BLEU score of 41.0 at less than one-quarter the training cost.



State-of-the-art BLEU scores on WMT 2014 translation tasks.